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CleanNode – Smart Public Toilet Monitoring System

Capstone Project

Course: BCA | Domain: Internet of Things

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Agenda

01 **Introduction**
Problem, solution and scope

02 **Background**
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03 **Requirements**
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05 **Implementation**
Web dashboard screens

06 **Testing**
15 test cases and results

07 **Conclusion**
Outcomes and future work

CleanNode at a Glance

A low-cost IoT and machine learning system that tells municipal authorities which public toilet needs cleaning, and when.

5
sensor inputs per node

0–100
Predictive Cleanliness Index

5 s
data update interval

4
user roles supported

7
Firebase data nodes

15/15
test cases passed

01

Introduction

Why public toilet maintenance needs real-time data



Problem • Current practice • Solution • Scope

Problem Statement

Public toilets are cleaned on fixed schedules, and their condition is known only when someone inspects them.

- Facilities can stay unusable for hours after they become dirty
- Lightly used toilets are cleaned as often as busy ones, wasting staff time
- Bad odour, water failure and overcrowding are noticed only after complaints
- Poor upkeep discourages people from using public toilets at all

Existing systems lack

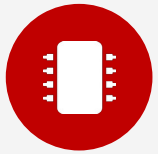
- Real-time monitoring
- Automated alerts
- Data-driven cleanliness evaluation
- Condition-based maintenance scheduling

Current Practice vs CleanNode

	Current practice	With CleanNode
Cleaning schedule	Fixed timetable	**Driven by live condition**
Problem detection	Inspection or complaint	**Automatic, within seconds**
Cleanliness assessment	Subjective, by eye	**PCI score from 0 to 100**
Flush failure	Found by the next user	**Occupancy without water flow flagged**
Staff allocation	Same effort everywhere	**Duties assigned where needed**
Records	Paper or none	**Full history in Firebase**

Proposed Solution

An IoT sensor node in every cubicle reports its condition continuously; a machine learning model turns the readings into a Predictive Cleanliness Index (PCI) that drives alerts and cleaning duties.



Sense

ESP32 reads gas, humidity, occupancy, water flow



Transmit

Readings published over MQTT



Analyse

Random Forest computes PCI; rules detect anomalies



Store

Results saved in Firebase



Act

Dashboard and app raise alerts and assign duties

Scope

In scope

- Public and community sanitation facilities with multiple cubicles
- Cleanliness monitoring with IoT sensors
- Real-time alerts and dashboard visualisation
- Rule-based logic and machine learning for PCI and anomaly detection
- Assignment and tracking of cleaning duties

Out of scope

- Robotic or automated cleaning
- Control of equipment inside the restroom

Automated flushing and sanitiser spraying are listed as future enhancements.

02

Background

Domain survey, existing systems and literature



Domain • Existing systems • 10 research papers • Gaps

Domain Survey: Overview and Challenges

Project domain

- **Smart Cities**
- **IoT Systems**
- **AI and Machine Learning**

Challenges

Lack of real-time monitoring

Condition is unknown between inspections

Manual inspection delays

Problems are found late or after complaints

Poor maintenance tracking

No record of who cleaned what and when

No predictive cleaning

Schedules ignore actual usage patterns

Existing Systems and Gap

Existing system	What it offers	Limitation
Manual inspection	Periodic visits by supervisors	Slow, labour-intensive, no records
Swachhata-MoHUA app	Complaint reporting with photo and GPS	Reactive; no automatic monitoring or verification
Indore smart sanitation dashboard	IoT sensors and alerts for authorities	Expensive and proprietary
Namma Toilet	Multi-sensor monitoring concept	Not deployed in a real environment
Air-quality based systems	Odour and gas detection	Cannot judge overall cleanliness or flush failure

CleanNode combines automatic multi-sensor monitoring, a single cleanliness score and duty management in a low-cost system.

Literature Survey: Ten Papers, 2020–2025

2020

- Deshmukh: IoT toilet monitoring
- Ha: sensor data fusion
- Pardini: citizen smart waste

2021

- Dineva: scalable AWS IoT
- Romano: gamification

2023

- Lokman: predictive toilet maintenance

2025

- Azman: restroom hygiene IoT
- Stephan: ProWaste ML
- Vibhute: smart drainage
- Zhang: real-time safety alerts

Three papers address toilet monitoring directly; the rest supply techniques reused in CleanNode.

Literature Survey (1/2)

Paper	Key contribution	Relevance / gap for CleanNode
Deshmukh et al., IEEE B-HTC 2020	IoT toilet monitoring with odour, usage, RFID and ARIMA forecasting	No combined cleanliness score
Azman et al., IEEE Access 2025	Ammonia and IAQ thresholds trigger condition-based cleaning	Air quality only; no flush or surface check
Lokman et al., IEEE Access 2023	ARIMA maintenance prediction with genetic-algorithm janitor scheduling	Tested in one building
Stephan et al., Scientific Reports 2025	ML predicts waste-centre criticality; Flutter app alerts crews	Waste domain; idea of proactive prioritising
Ha et al., IEEE Sensors J. 2020	Kalman-filter fusion of IAQ and humidex into one index	Fusion not applied to toilet cleanliness

Literature Survey (2/2)

Paper	Key contribution	Relevance / gap for CleanNode
Vibhute et al., IJFMR 2025	Flow, temperature and gas sensors detect drainage blockages	Flow sensing idea reused for flush health
Zhang et al., Scientific Reports 2025	Edge and cloud alerts with sub-450 ms latency	General safety; not sanitation
Dineva and Atanasova, Animals 2021	Scalable AWS IoT architecture for 10,000 devices/s	Shows cloud scaling for city-wide nodes
Pardini et al., Sensors 2020	Citizen-facing smart waste bins with web and mobile apps	Citizen access idea reused for public app
Romano et al., JAIHC 2021	Gamification raised civic participation in a field trial	Not yet used in sanitation apps

Research Gaps Addressed by CleanNode

Flush health

Occupancy without water flow is detected as a flush issue

Single cleanliness score

Readings combined into a PCI from 0 to 100

Usage-based scheduling

Usage patterns and alerts drive cleaning duties

Urgent alerts

Danger Zone alerts with 15-minute escalation

Citizen access

Public app with toilet map and feedback

Scale

Multiple facilities and cubicles on one cloud backend

03

Requirements

What CleanNode must do, and how well



10 functional • 7 non-functional

Functional Requirements

- **FR1** Real-time monitoring of toilet sensor data
- **FR2** Management of multiple facilities and cubicles
- **FR3** Monitoring of temperature, humidity, gas, occupancy and flush status
- **FR4** Calculation of the Predictive Cleanliness Index (PCI)
- **FR5** Alerts for abnormal conditions
- **FR6** Prediction of cubicles requiring maintenance
- **FR7** Detection of multi-sensor anomalies
- **FR8** Analysis of toilet usage patterns
- **FR9** Assignment and reassignment of cleaning duties
- **FR10** History of sensor readings, alerts and maintenance

Non-Functional Requirements

Performance

Real-time updates with quick response

Reliability

Continuous and accurate monitoring

Scalability

Multiple facilities and cubicles

Security

Secure, role-based admin and staff access

Usability

Simple, easy-to-use dashboard

Maintainability

Modular design; limits set from Settings

Availability

Monitoring and alerts always reachable

04

System Design

Technology, architecture, diagrams and data



Stack • Architecture • Use cases • Hardware • AI/ML • Database

Technology Stack

Hardware edge

ESP32 DevKit V1, MQ-135, DHT22, PIR, YF-S201

Telemetry

MQTT over HiveMQ Cloud (topic pes/cleannode/iot)

Backend and AI/ML

FastAPI on Python 3.12, Scikit-Learn Random Forest, Pandas, NumPy

Data

Firebase Realtime Database and Firebase Authentication

Web dashboard

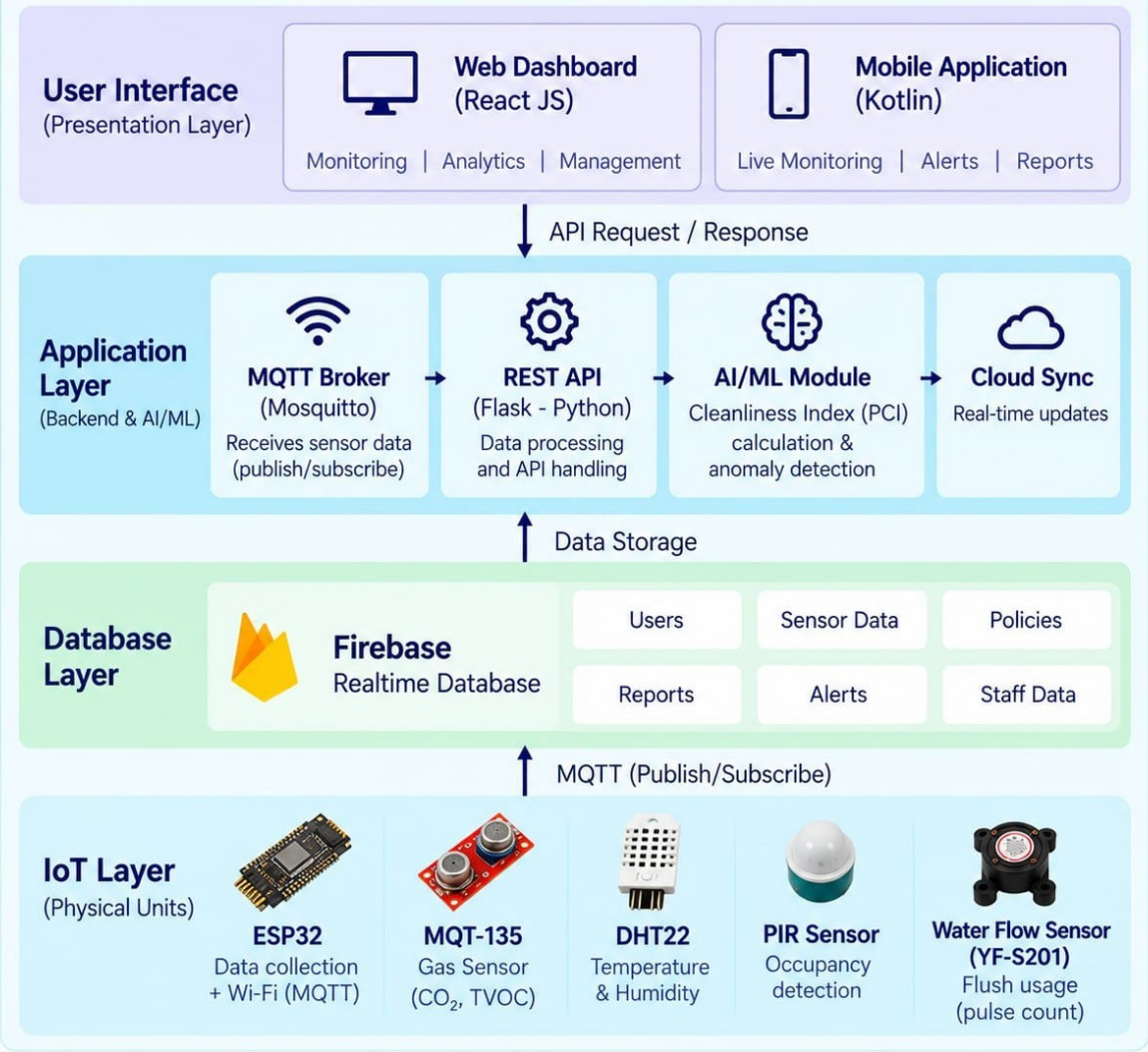
React 18, Vite, Tailwind CSS

Mobile app

Kotlin, Jetpack Compose (Android 14), Room, CameraX

Tools: Arduino IDE, VS Code, Android Studio, Firebase Console, Postman, Git

System Architecture



User interface

React web dashboard and Kotlin mobile app

Application

MQTT broker, REST API, AI/ML module, cloud sync

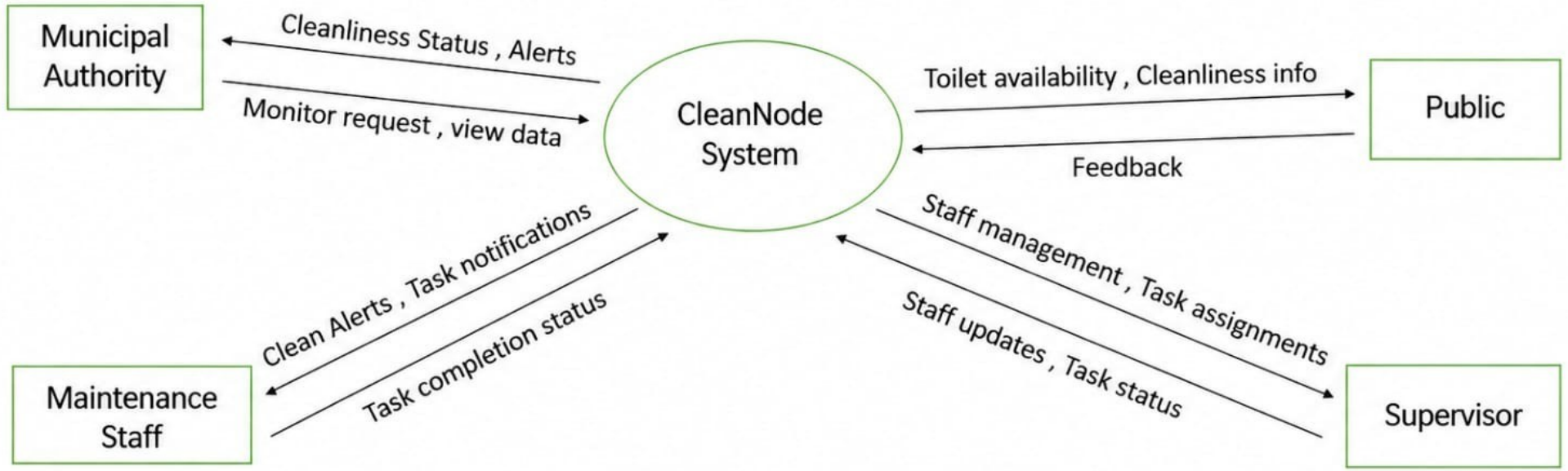
Database

Firebase Realtime Database

IoT

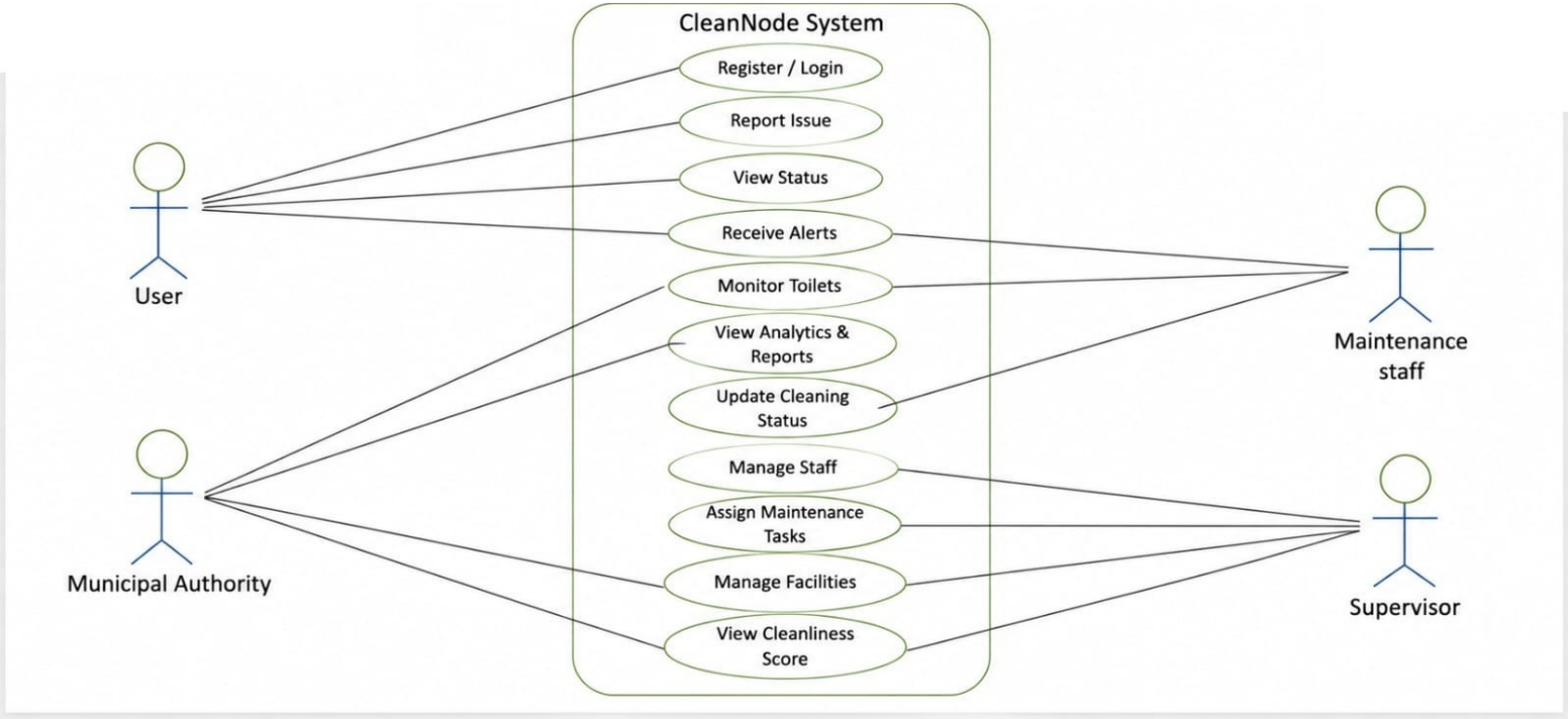
ESP32 with gas, temperature, humidity, occupancy and flow sensors

Context Flow Diagram



Four external entities exchange information with CleanNode: the Municipal Authority, Maintenance Staff, the Public and the Supervisor.

Use Case Diagram



User (public)

Register, report issues, view status, receive alerts

Municipal Authority

Monitor toilets, analytics, facilities, cleanliness score

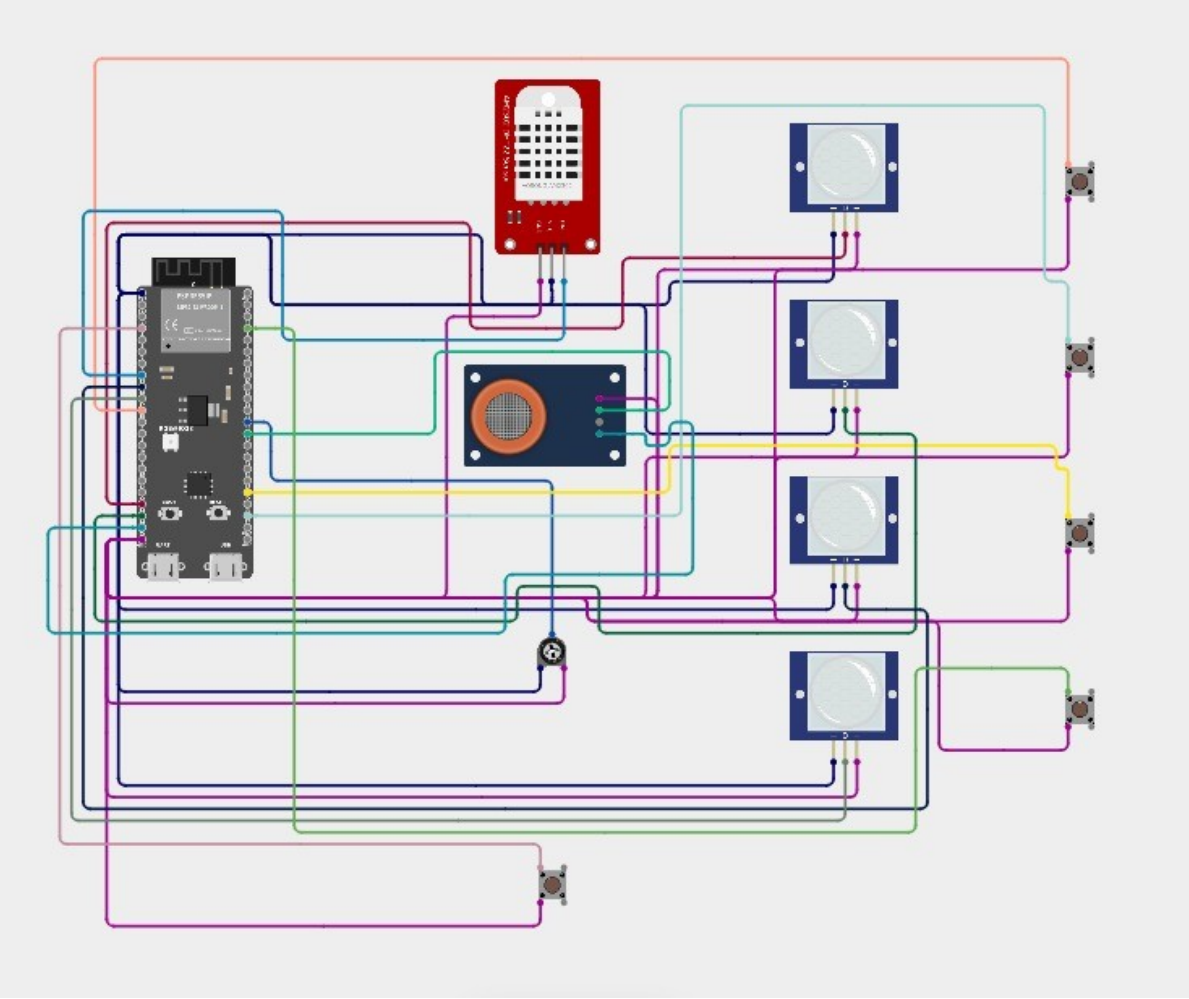
Maintenance Staff

Receive alerts, update cleaning status

Supervisor

Manage staff, assign tasks, view scores

Hardware Integration



ESP32	Central controller; sends data over Wi-Fi/MQTT
MQ-135	Gas / ammonia level and bad smell
DHT22	Temperature and humidity
PIR	Occupancy in each cubicle
YF-S201	Water flow and flush activity
Push buttons	Simulate sensor inputs during testing

AI/ML Module: Cleanliness Index and Alerts

1

Fuse readings

Gas, humidity, temperature, occupancy and water flow

2

Predict PCI

Random Forest regression gives a score from 0 to 100

3

Debounce

Condition must hold for 3 consecutive readings

4

Alert and escalate

Unresolved critical issues escalate after 15 minutes

Alert types

Bad Smell

Gas level above the configured limit

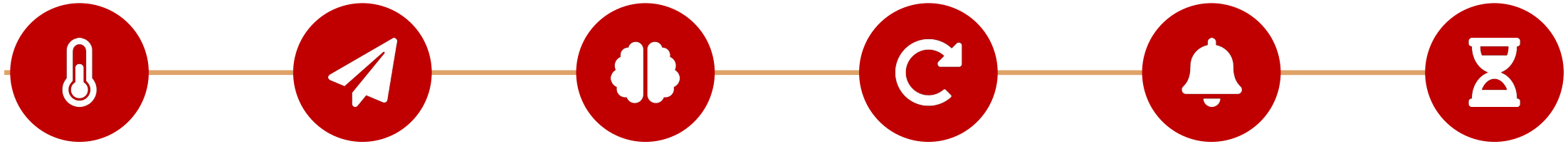
Flush Issue

Occupancy detected but no water flow

Danger Zone

PCI below the critical limit

From Reading to Action



Every 2 s
DHT22 sampled

Every 5 s
ESP32 publishes JSON over MQTT

On arrival
PCI computed, anomalies checked

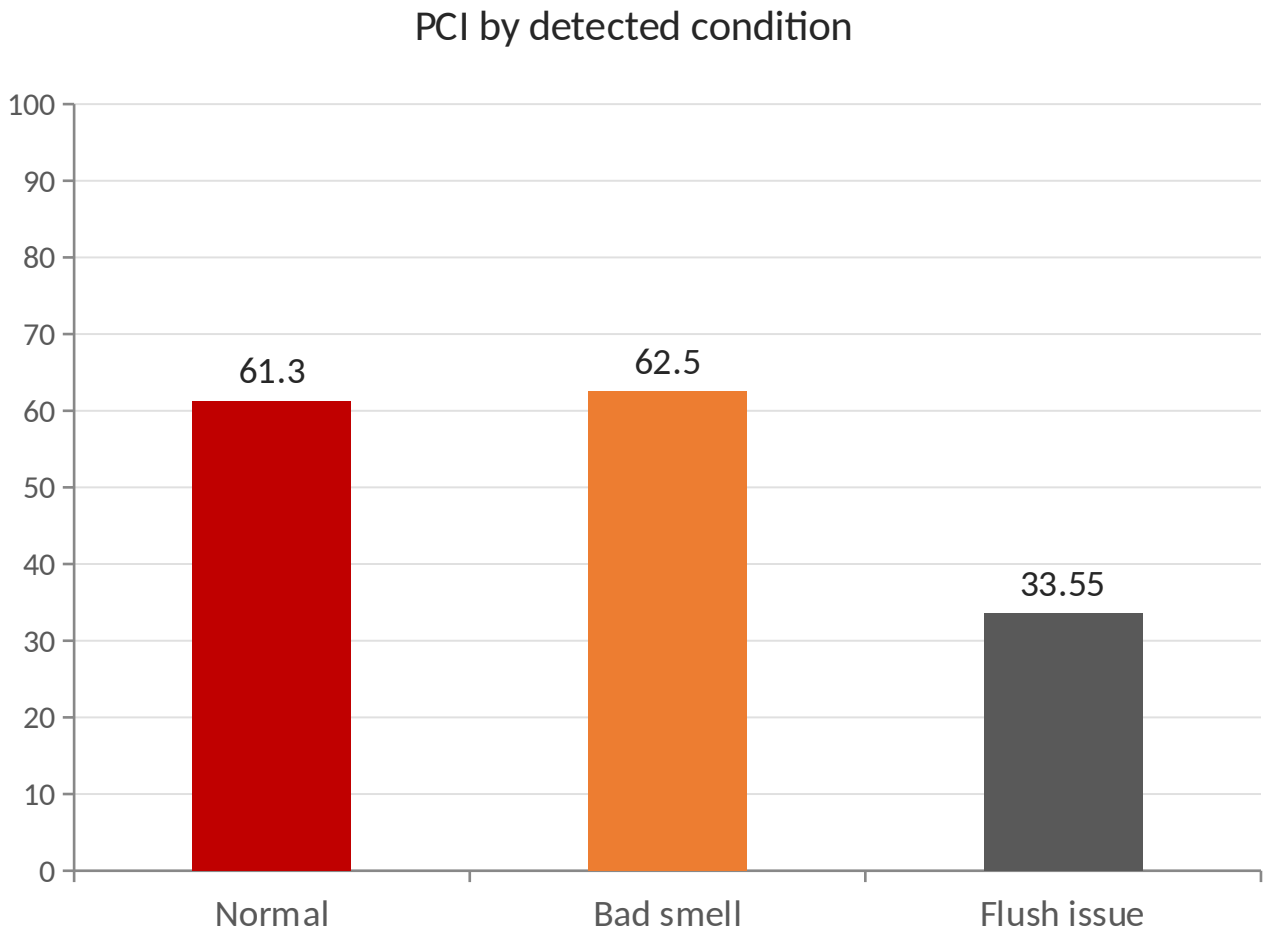
3 readings
Condition confirmed (debounce)

Alert
Shown on dashboard and app

15 min
Unresolved critical issue escalated

Debouncing over three readings means a single noisy value cannot raise a false alarm, while a genuine problem is still reported within about 15 seconds.

Sample Readings and PCI



Normal

Gas 390, humidity 46%, occupied, water flowing

Bad smell

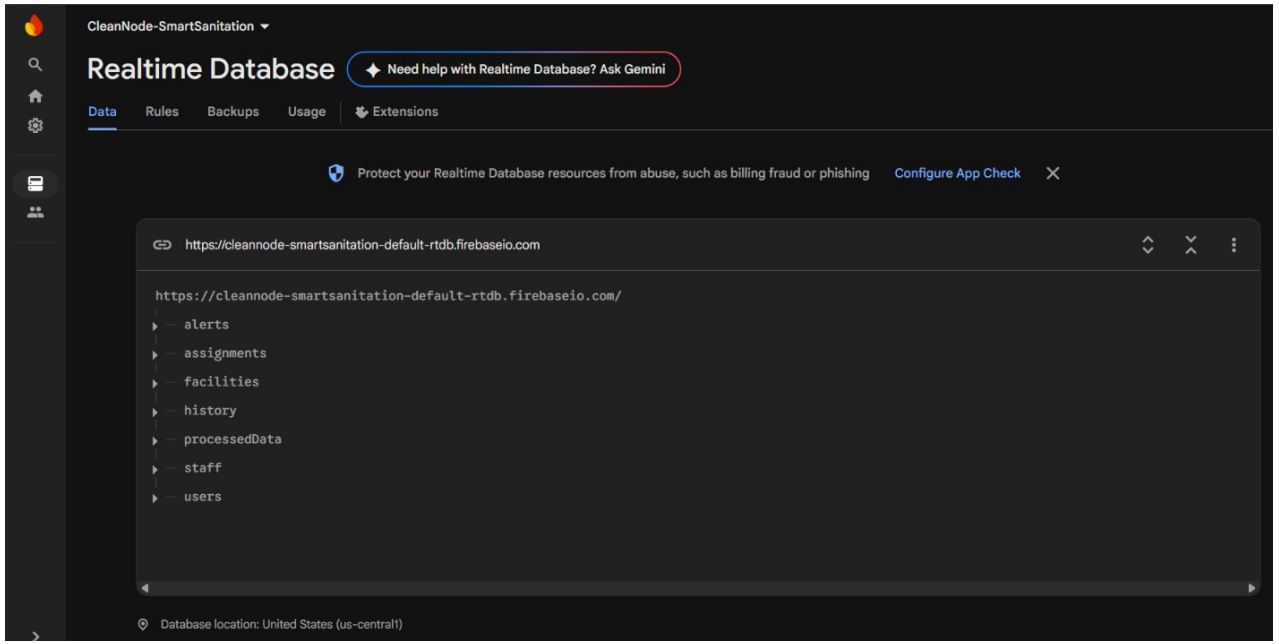
Gas 494 above the limit; vacant

Flush issue

3 occupants, no water flow; PCI drops to 33.55

Bad smell is an air-quality alert, so the overall PCI can still be moderate; a flush failure lowers the PCI sharply.

Database: Firebase Realtime Database



`facilities`	Live readings: facility → block → cubicle
`alerts`	Active Flush Issue, Bad Smell, Danger Zone
`assignments`	Cleaning duties assigned to staff
`history`	Time series of sensor readings
`processedData`	Computed PCI and cleanliness scores
`staff`	Staff profiles and roles
`users`	Admin / Staff / Public role mapping

05

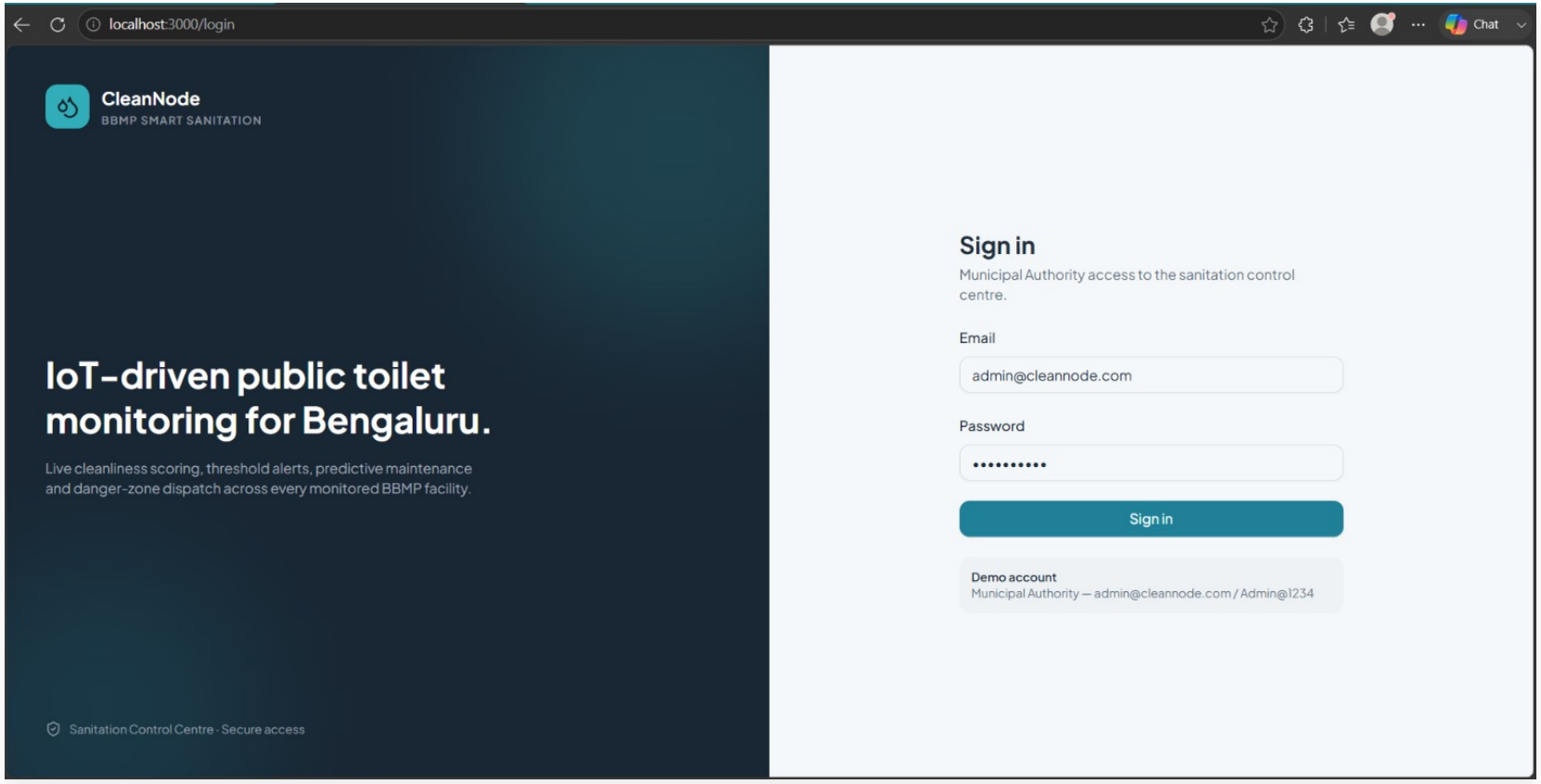
Implementation

The CleanNode web dashboard



Login • Overview • Facilities • Live monitor • Anomalies • Duties

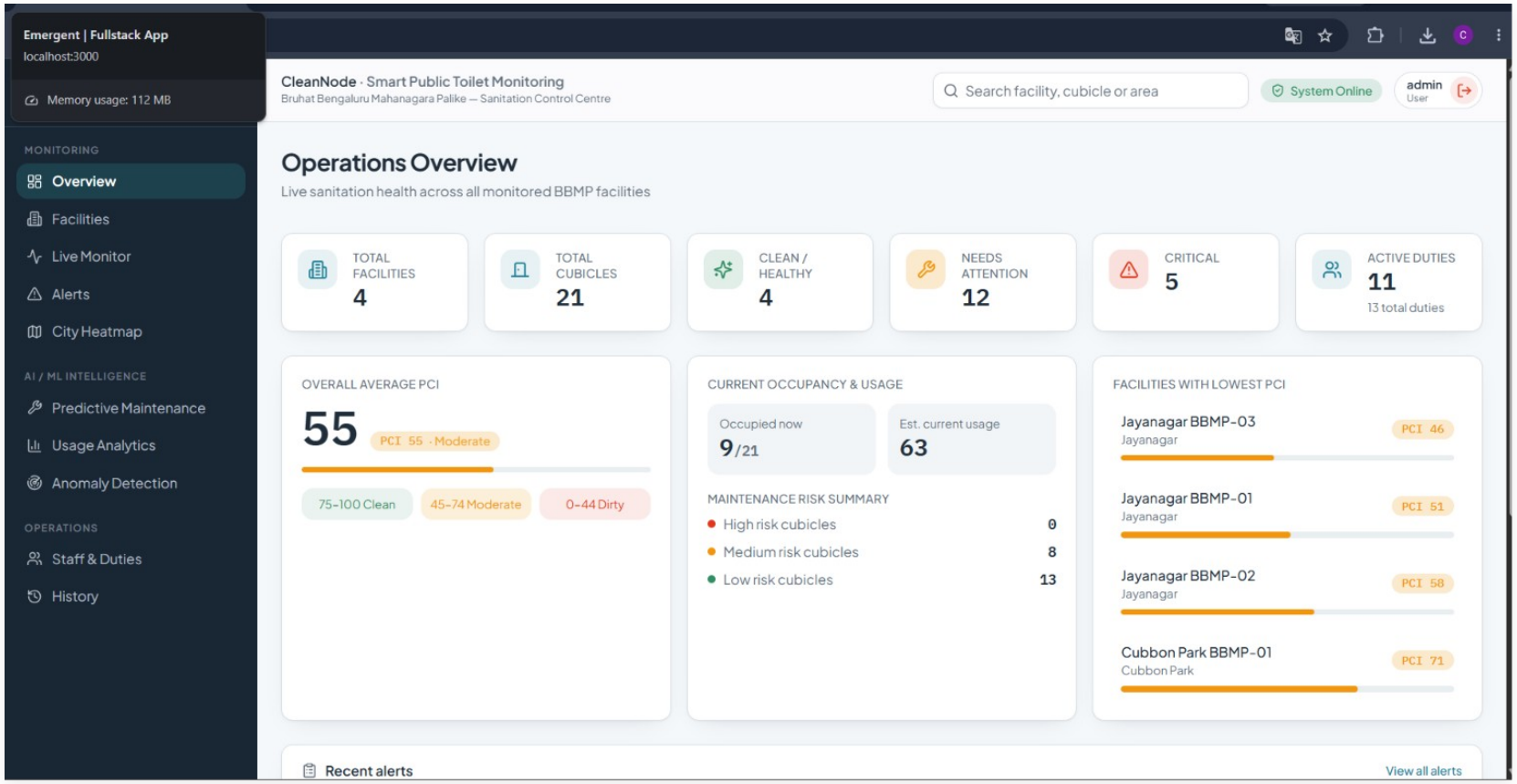
Web Dashboard: Login and Authentication



Highlights

- Secure entry for authority and staff users
- Firebase Authentication verifies each user and role
- Only signed-in users reach management functions

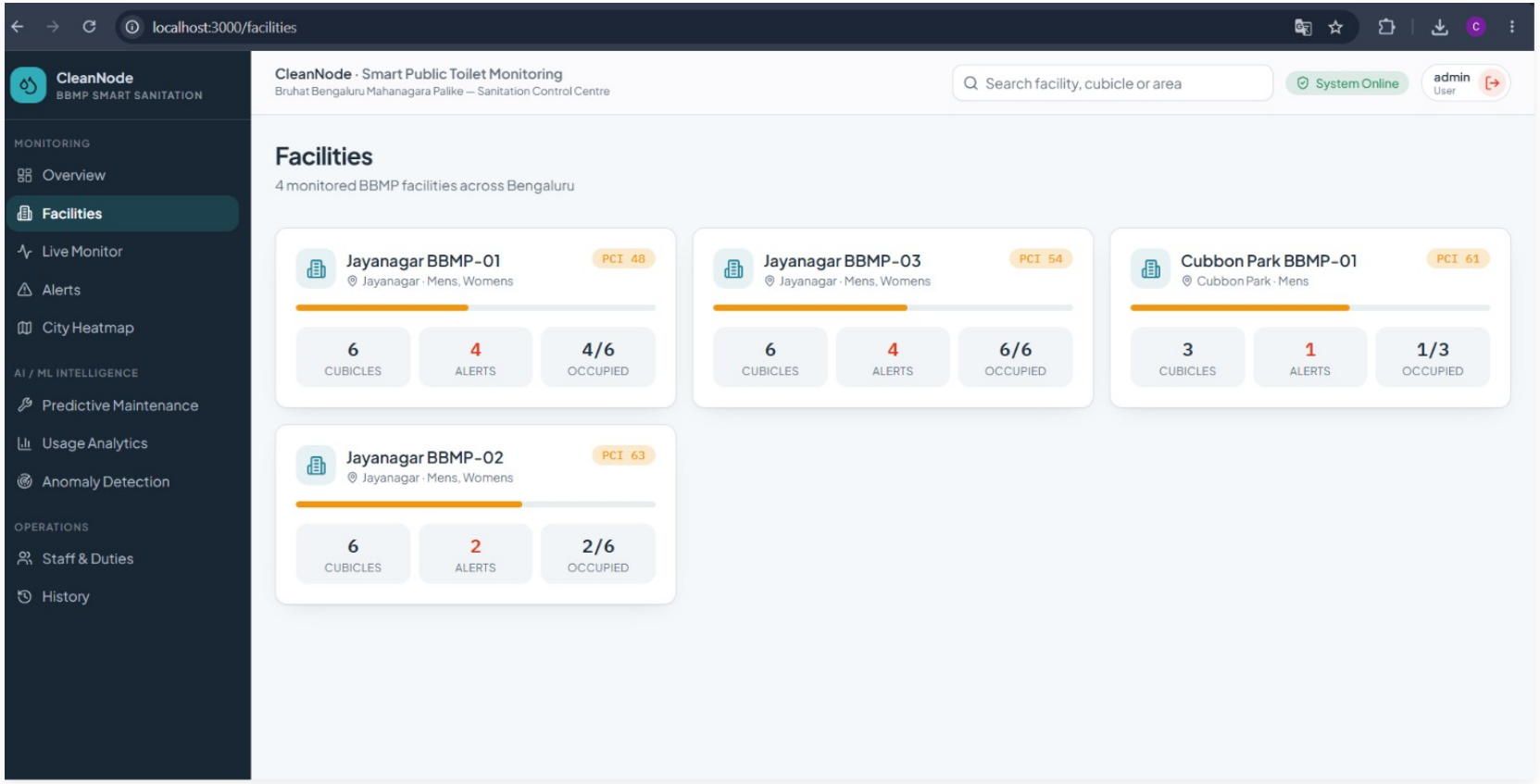
Web Dashboard: Operations Overview



Highlights

- All facilities and cubicles in one view
- Clean, attention and critical counts
- Overall PCI, occupancy, usage and active duties

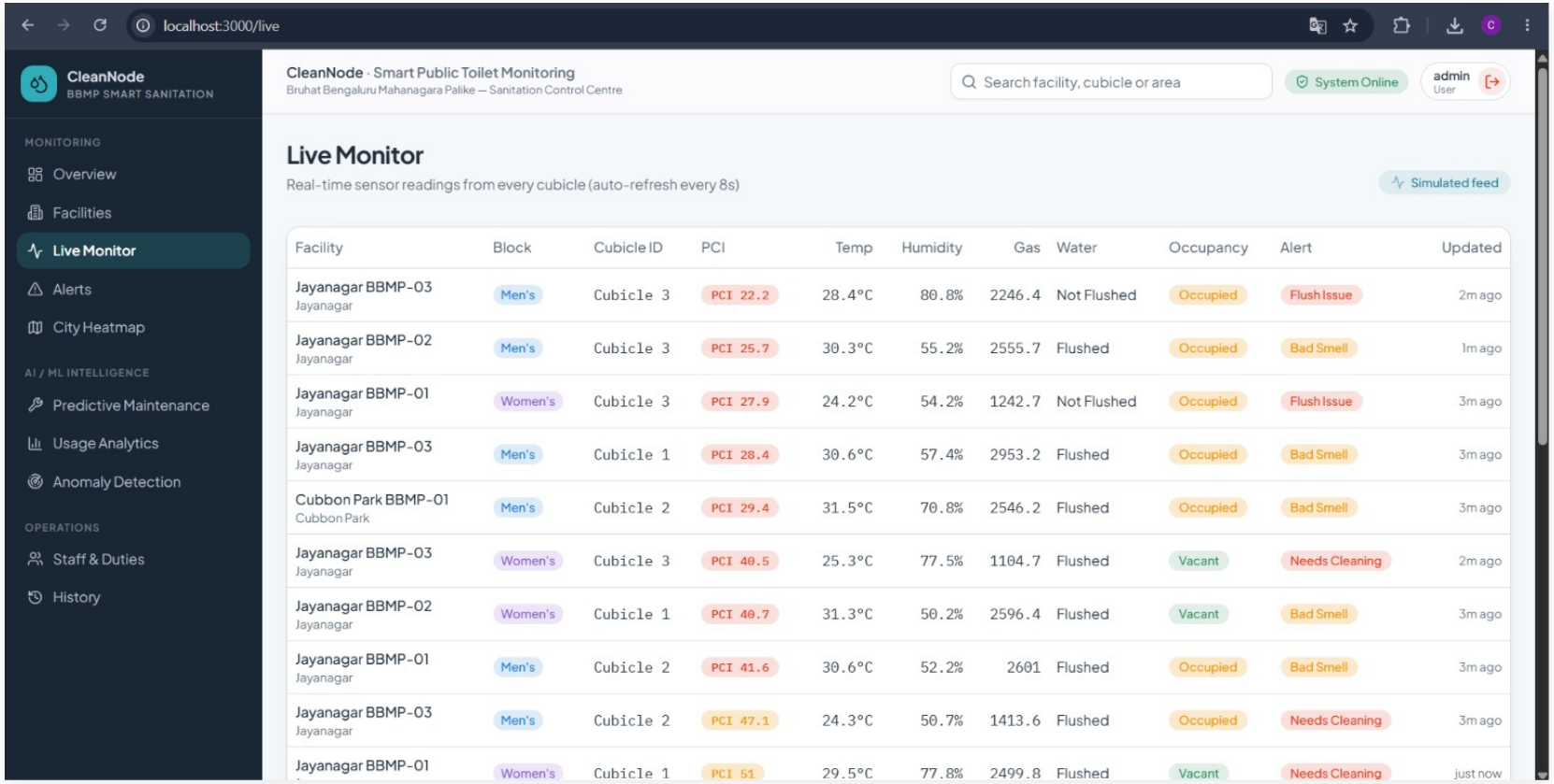
Web Dashboard: Facilities



Highlights

- Each BBMP facility with its PCI
- Cubicle count, alerts and occupancy
- Compare conditions across locations

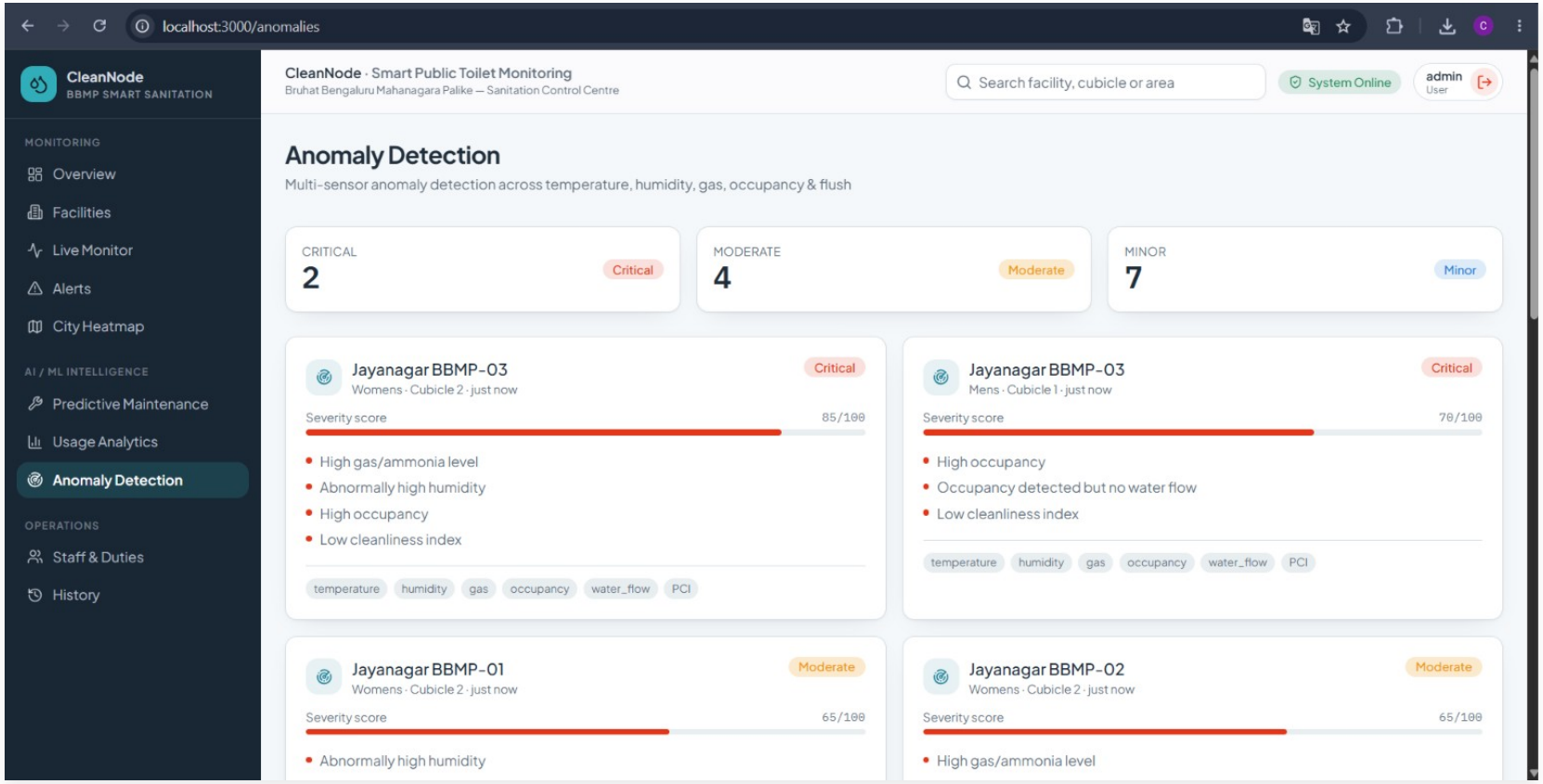
Web Dashboard: Live Monitor



Highlights

- Latest readings for every cubicle
- Temperature, humidity, gas, flush, occupancy, PCI
- Refreshes automatically from Firebase

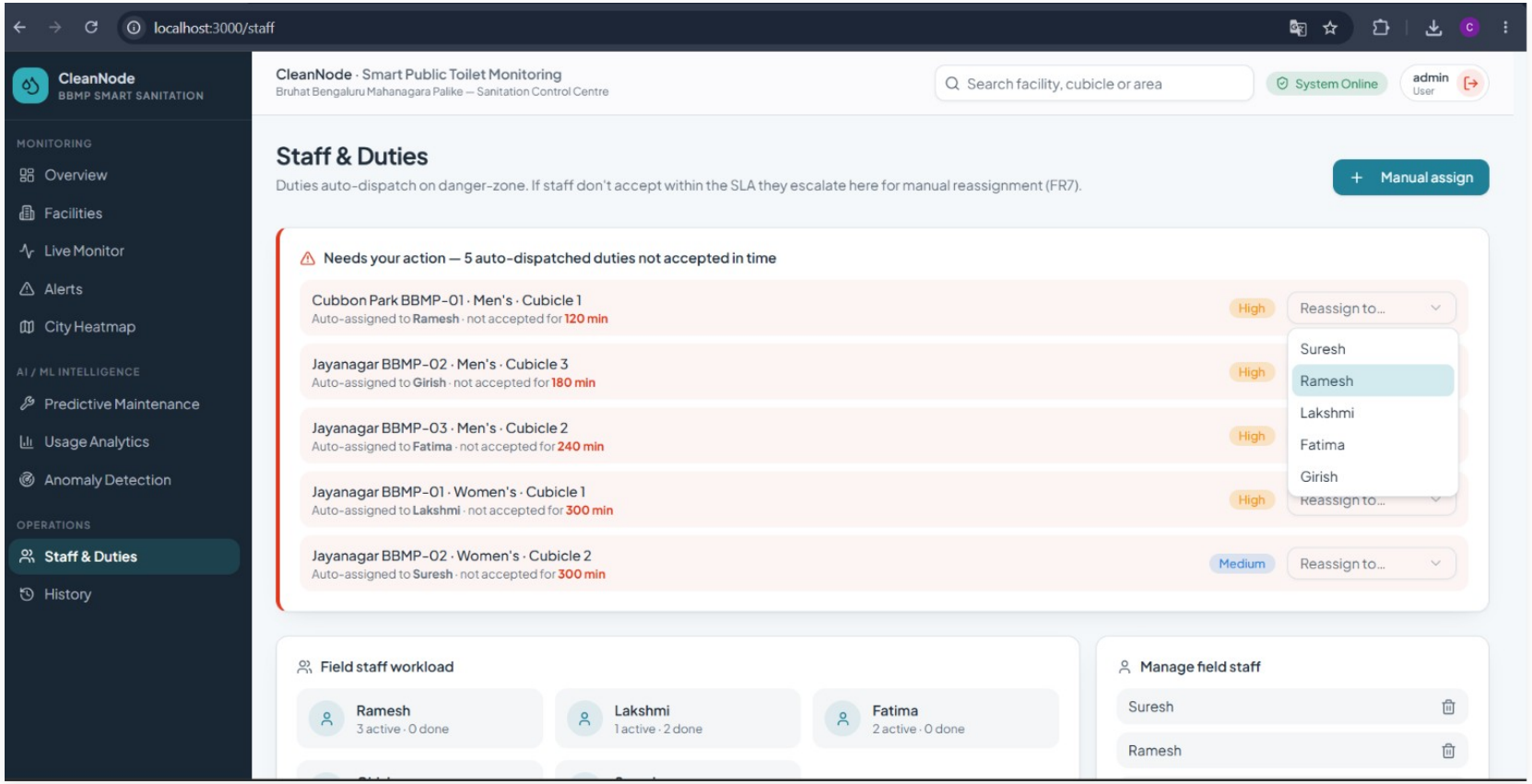
Web Dashboard: Anomaly Detection



Highlights

- Unusual combinations of sensor conditions
- Severity: Critical, Moderate or Minor
- Combines several sensors, not just one

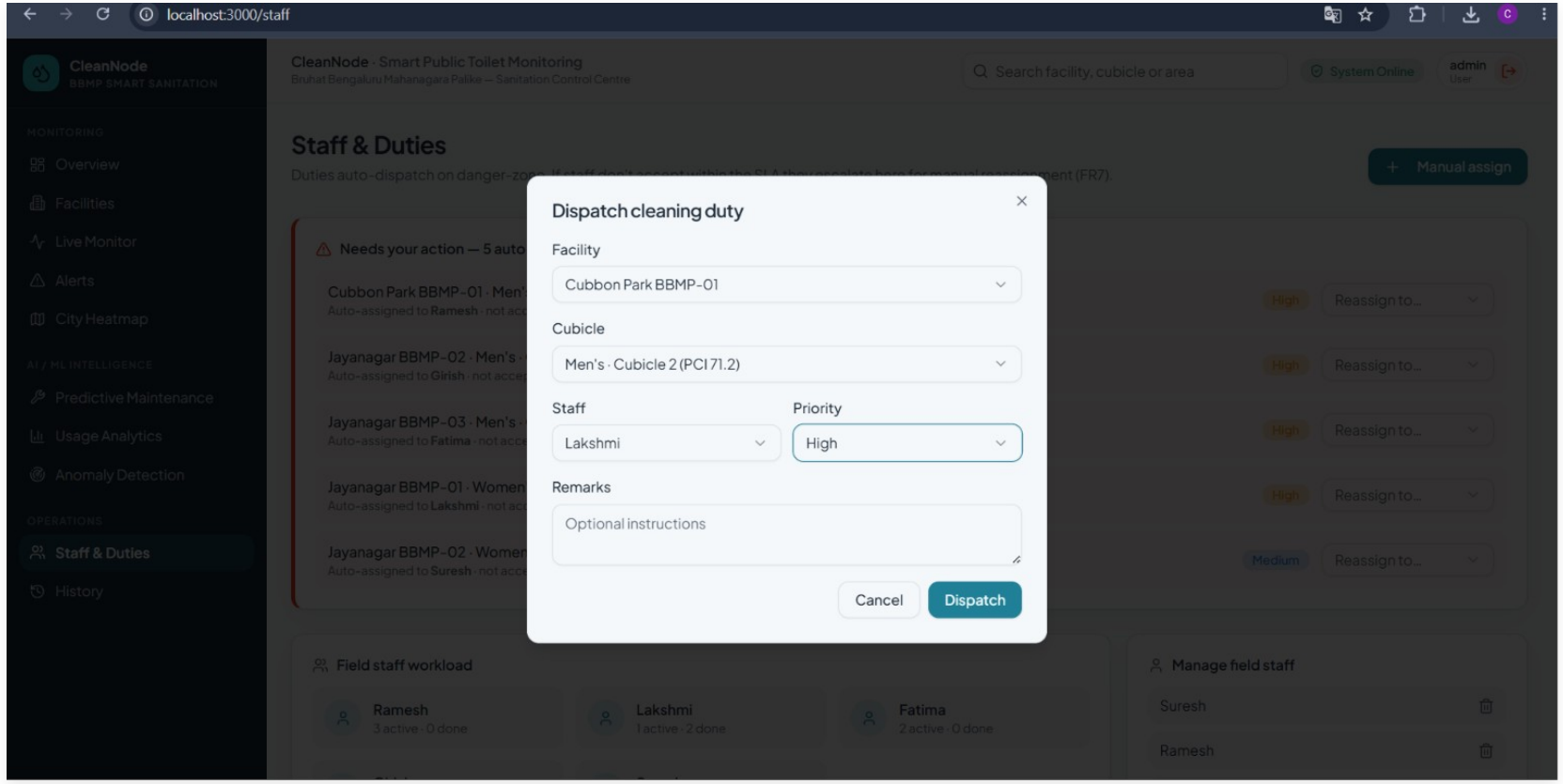
Web Dashboard: Staff and Duties



Highlights

- Assigned duties, workload and priority
- Duties that need reassignment are flagged
- Assign or move duties between staff

Web Dashboard: Dispatch Cleaning Duty



Highlights

- Select facility, cubicle, staff and priority
- Add remarks before dispatching
- Reassign if the staff member is unavailable

06

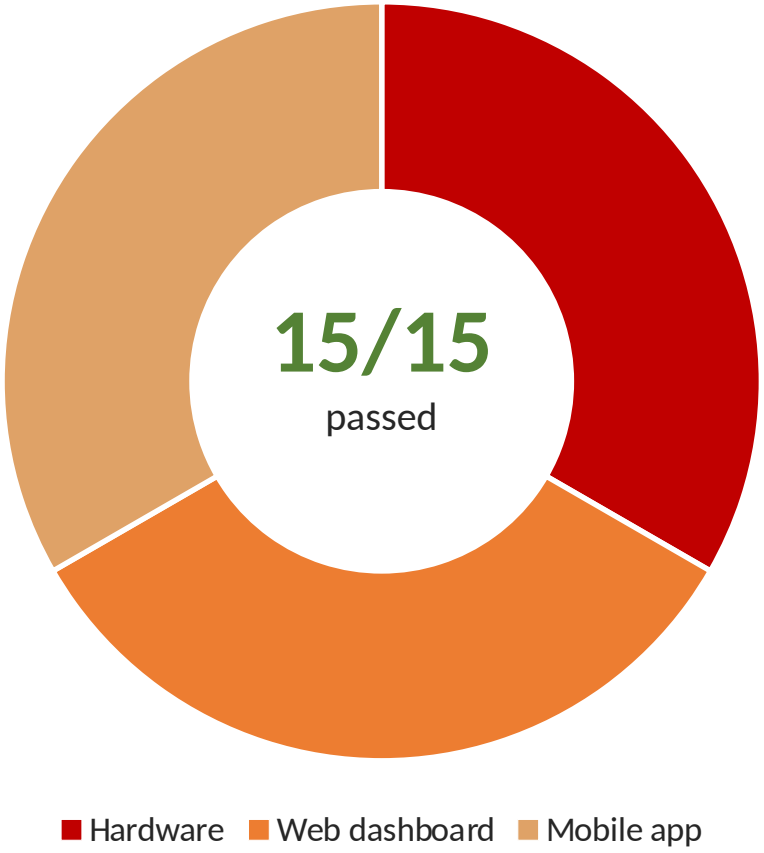
Testing

Hardware, web dashboard and mobile application



15 test cases • Sensor validation

Testing Summary



Hardware and sensors

5 tests: MQ-135, PIR, YF-S201, DHT22, ESP32 data push

Web dashboard

5 tests: login, ward filter, live update, settings, escalation

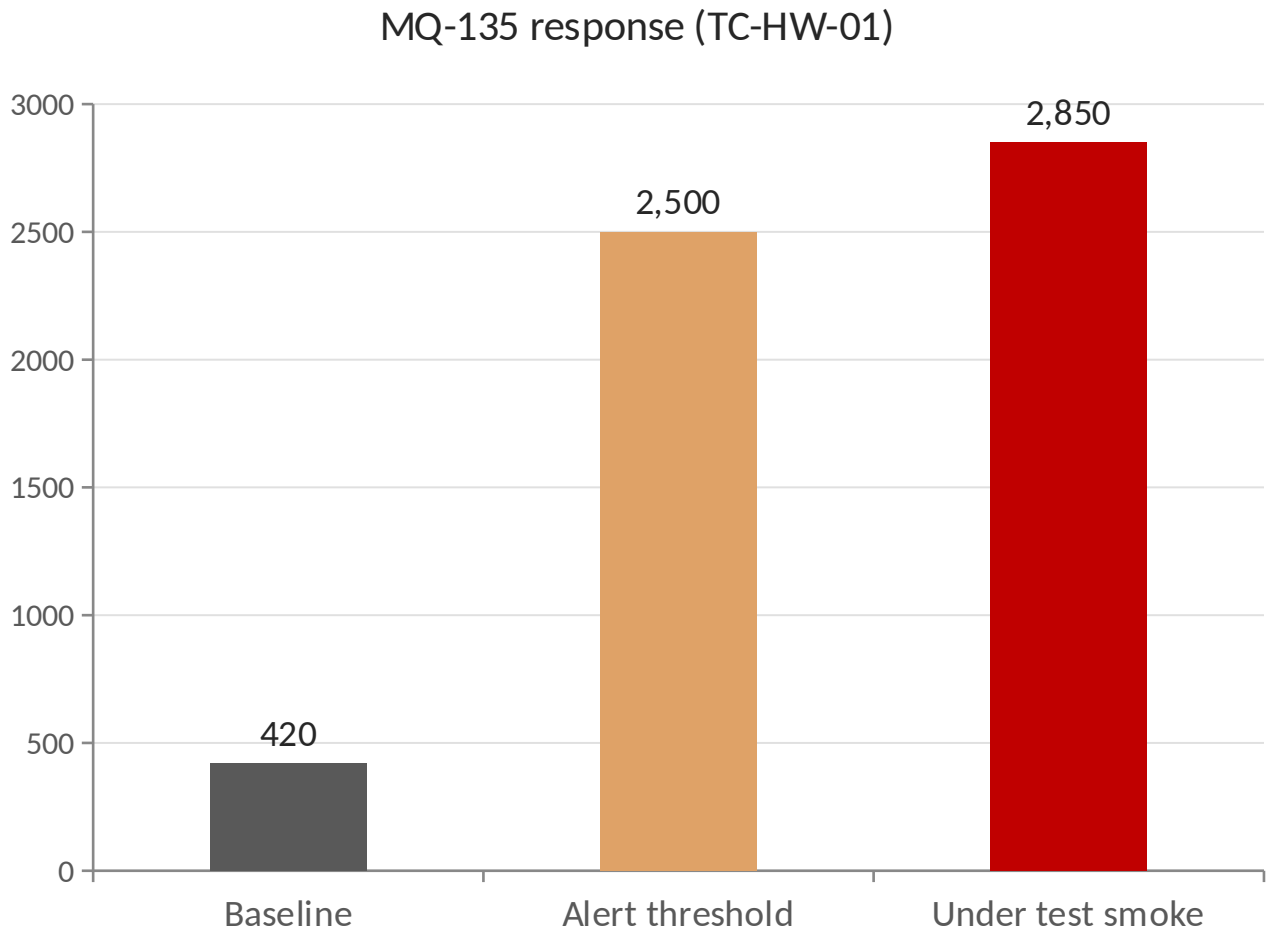
Mobile app

5 tests: KPIs, spike demo, alerts, map, citizen feedback

Test Cases: Hardware and Sensors

ID	Module	Test condition	Expected	Observed	Status
TC-HW-01	Gas (MQ-135)	Exposed to test smoke / foul air	Reading exceeds 2500 ppm	Rose from 420 to 2850 ppm	Pass
TC-HW-02	Motion (PIR)	Hand wave / person moving	Signal goes LOW to HIGH	Status changed to Occupied	Pass
TC-HW-03	Flow (YF-S201)	Water flows during flush	Pulses counted (F = 7.5Q)	3.5 litres measured	Pass
TC-HW-04	DHT22	Continuous room sampling	Reading every 2 s	26.8 °C and 62% RH recorded	Pass
TC-HW-05	ESP32 push	Wi-Fi send to MQTT / cloud	280-byte JSON every 5 s	Delivered with no drops	Pass

Sensor Validation Highlights



2850 ppm

gas peak, above the 2500 ppm threshold

3.5 L

water measured by YF-S201

26.8 °C / 62%

temperature and humidity (DHT22)

0

packets dropped from ESP32

Test Cases: Web Dashboard

ID	Module	Test condition	Expected	Observed	Status
TC-WEB-01	Admin login	Log in via Firebase	Admin verified and logged in	Role verified, login worked	Pass
TC-WEB-02	Ward filter	Switch ward 153 to 111	Data changes without refresh	Ward changed instantly	Pass
TC-WEB-03	Live update	Change a value in Firebase	New value shown immediately	Appeared automatically	Pass
TC-WEB-04	Settings	Change cleanliness limit	Status follows new limit	Updated without restart	Pass
TC-WEB-05	Escalation	Critical issue open 15 min	Alert shown	Alert shown and recorded	Pass

Test Cases: Mobile Application

ID	Module	Test condition	Expected	Observed	Status
TC-MOB-01	KPI overview	Launch app on Firebase	Live KPIs (5 monitored, PCI 75)	All four cards populated	Pass
TC-MOB-02	Spike demo	Tap Trigger Spike	PCI drops into Danger Zone	Danger Zone = 1, Critical	Pass
TC-MOB-03	Alerts	Open Alerts after spike	Alert with time and cubicle	Shown for Jayanagar e-Toilet	Pass
TC-MOB-04	Restroom map	Open Map tab	BBMP toilets on map	Markers with ward names	Pass
TC-MOB-05	Feedback	Submit 4/5 rating and comment	Saved to database	Toast shown; stored in Firebase	Pass

07

Conclusion

Outcomes and future enhancements



Outcomes • Future work • References

Conclusion

CleanNode replaces fixed cleaning schedules with maintenance driven by the actual condition of each toilet.

Sensing

ESP32 nodes with gas, climate, occupancy and flow sensors stream data over MQTT

Intelligence

Random Forest PCI with debouncing and multi-sensor anomaly detection

Action

Web dashboard and Android app link every alert to a cleaning duty

Verified

All 15 hardware, web and mobile test cases passed

Future Enhancements

Automated flush and sanitiser

Solenoid valves triggered by ammonia spikes

Solar power

Rooftop panels with LiFePO4 batteries for each node

Consumables monitoring

Ultrasonic levels for water tanks and soap

Fault and tamper detection

Alerts for cut wires, covered sensors or damage

Geofenced staff alerts

Phone alert within 100 m of a Danger Zone toilet

References

1. P. Deshmukh et al., Intelligent Public Toilet Monitoring System Using IoT, IEEE B-HTC, 2020
2. F. I. Azman et al., An IoT-Based Hygiene Monitoring System in the Restroom, IEEE Access, Vol. 13, 2025
3. A. Lokman, R. K. Ramasamy, C.-Y. Ting, Scheduling and Predictive Maintenance for Smart Toilet, IEEE Access, Vol. 11, 2023
4. T. Stephan et al., ProWaste for Proactive Urban Waste Management Using IoT and ML, Scientific Reports, Vol. 15, 2025
5. Q. P. Ha, S. Metia, M. D. Phung, Sensing Data Fusion for Enhanced Indoor Air Quality Monitoring, IEEE Sensors Journal, Vol. 20, 2020
6. M. Vibhute et al., Smart Drainage Monitoring System, IJFMR, Vol. 7, 2025
7. H. Zhang, R. Zhang, J. Sun, Real-Time IoT-Based Public Safety Alert and Emergency Response Systems, Scientific Reports, Vol. 15, 2025
8. K. Dineva, T. Atanasova, Design of Scalable IoT Architecture Based on AWS for Smart Livestock, Animals, Vol. 11, 2021
9. K. Pardini et al., A Smart Waste Management Solution Geared Towards Citizens, Sensors, Vol. 20, 2020
10. M. Romano, P. Díaz, I. Aedo, Gamification-less: May Gamification Really Foster Civic Participation?, JAIHC, 2021

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